Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

Aim:- To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.

Theory :-

Histogram enhancement is a technique in Digital Image Processing used to improve the contrast and clarity of an image by modifying its pixel intensity distribution. Histogram Equalization automatically redistributes the intensity values over a wider range, which can improve the overall contrast and make hidden details more visible. Histogram Matching, also called histogram specification, modifies an image so that its histogram becomes similar to a selected reference image. Unlike equalization, which uses a general distribution, histogram matching allows the output appearance to be controlled according to the chosen reference. Both techniques can improve image quality, but their effectiveness depends on the original image and the desired enhancement. In this experiment, both methods are applied to the same degraded images and their results are compared based on contrast, clarity, and visibility of important details.

Image 1 — Histogram Equalization and Histogram Matching

Explanation

The program initially loads the photo as a grayscale image and verifies that it has been loaded properly. A reference image with a Gaussian distribution is then generated using a mean of 128 and a standard deviation of 50. After that, histogram equalization is performed with `cv2.equalizeHist()` to enhance the overall contrast of the image. Histogram matching is then carried out using `match_histograms()` to make the intensity distribution of the original image similar to that of the generated reference image. Finally, the original, equalized, and histogram-matched images are displayed along with their respective histograms. The program also computes the mean and standard deviation of all three images for comparison.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
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Code:-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.exposure import match_histograms


# Read image in grayscale
src_image = cv2.imread("ex1_1.png", cv2.IMREAD_GRAYSCALE)


if src_image is None:
    print("Image not found!")
    exit()


# -------------------------
# Reference image for matching (Gaussian target, mean=128, std=50)
# -------------------------
np.random.seed(42)
reference_image = np.random.normal(loc=128, scale=50, size=(300, 300))
reference_image = np.clip(reference_image, 0, 255).astype(np.uint8)


# -------------------------
# Histogram Equalization
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

```
# -------------------------

equalized_image = cv2.equalizeHist(src_image)


# -------------------------

# Histogram Matching

# -------------------------

matched_image = match_histograms(src_image, reference_image)

matched_image = np.uint8(np.clip(matched_image, 0, 255))


# -------------------------

# Display Images

# -------------------------

plt.figure(figsize=(15, 5))


plt.subplot(1, 3, 1)

plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)

plt.title("Original")

plt.axis('off')


plt.subplot(1, 3, 2)

plt.imshow(equalized_image, cmap='gray', vmin=0, vmax=255)

plt.title("Histogram Equalized")

plt.axis('off')
```

```
plt.subplot(1, 3, 3)
plt.imshow(matched_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Matched")
plt.axis('off')

plt.show()

# -------------------------
# Histograms
# -------------------------
plt.figure(figsize=(15, 4))

plt.subplot(1, 3, 1)
plt.hist(src_image.ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")

plt.subplot(1, 3, 2)
plt.hist(equalized_image.ravel(), bins=256, range=(0, 255))
plt.title("Equalized Histogram")

plt.subplot(1, 3, 3)
plt.hist(matched_image.ravel(), bins=256, range=(0, 255))
```

```
plt.title("Matched Histogram")


plt.show()


# ------------------------

# Stats

# ------------------------

print(f"Original : mean={src_image.mean():.1f}  std={src_image.std():.1f}")

print(f"Equalized: mean={equalized_image.mean():.1f}  std={equalized_image.std():.1f}")

print(f"Matched  : mean={matched_image.mean():.1f}  std={matched_image.std():.1f}")
```

Expected Output

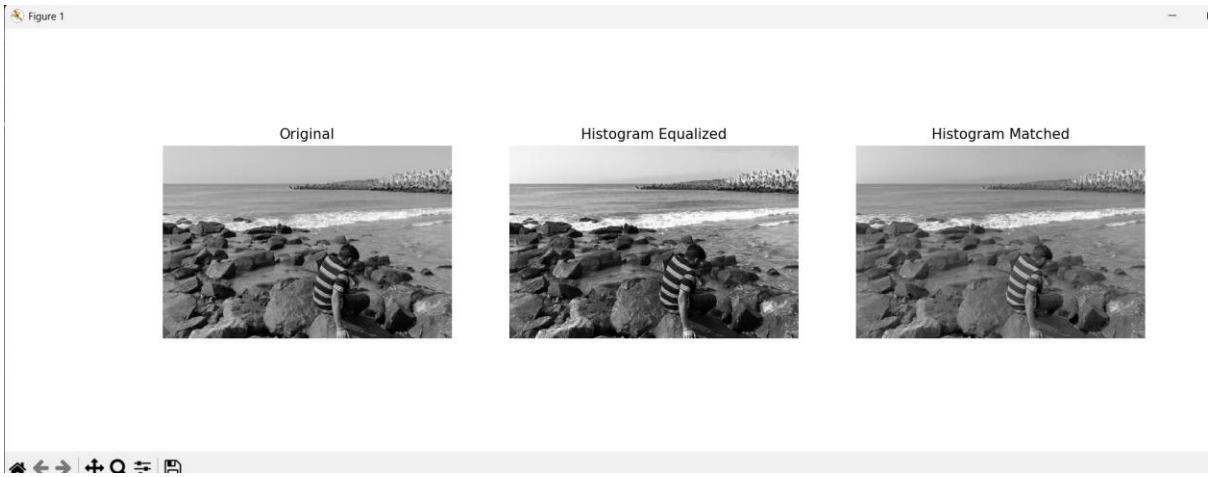
The output shows three images: Original, Histogram Equalized, and Histogram Matched. Three corresponding histograms are also displayed. At the end, the program prints the mean and standard deviation values for the original, equalized, and matched images.

Observation

The histogram-equalized image provides better overall contrast by spreading the pixel intensity values more evenly. The histogram-matched image has an intensity distribution that is closer to the chosen Gaussian reference image, resulting in a different brightness and contrast. The histograms clearly illustrate the changes made to the original intensity distribution by both methods. The calculated mean and standard deviation values also give a numerical comparison of the changes in brightness and contrast.

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

Output :-



Histograms:-

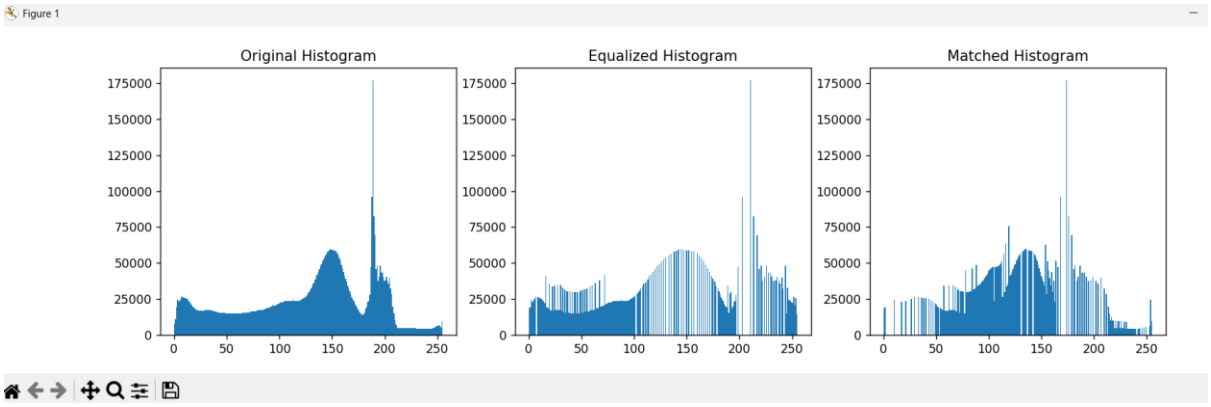


Image 2 — Circular Texture (Histogram Equalization + Matching):-

Explanation

The program loads the image as a grayscale image and verifies that it has been loaded successfully. A Gaussian reference image is created with a mean of 128 and a standard deviation of 50. The original image is then improved using histogram equalization, while

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

histogram matching modifies its intensity distribution to resemble the generated reference image. The original, equalized, and matched images are displayed along with their corresponding histograms. Finally, the mean and standard deviation of each image are calculated to compare their intensity distributions.

Code :-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

from skimage.exposure import match_histograms


src_image = cv2.imread("ex1_2.png", cv2.IMREAD_GRAYSCALE)


if src_image is None:

    print("Image not found!")

    exit()


np.random.seed(42)

reference_image = np.random.normal(loc=128, scale=50, size=(300, 300))

reference_image = np.clip(reference_image, 0, 255).astype(np.uint8)


equalized_image = cv2.equalizeHist(src_image)


matched_image = match_histograms(src_image, reference_image)

matched_image = np.uint8(np.clip(matched_image, 0, 255))


plt.figure(figsize=(15, 5))
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

```
plt.subplot(1, 3, 1)
plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)
plt.title("Original")
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(equalized_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Equalized")
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(matched_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Matched")
plt.axis('off')

plt.show()

plt.figure(figsize=(15, 4))

plt.subplot(1, 3, 1)
plt.hist(src_image.ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")

plt.subplot(1, 3, 2)
```

```
plt.hist(equalized_image.ravel(), bins=256, range=(0, 255))
plt.title("Equalized Histogram")

plt.subplot(1, 3, 3)
plt.hist(matched_image.ravel(), bins=256, range=(0, 255))
plt.title("Matched Histogram")

plt.show()

print(f"Original : mean={src_image.mean():.1f}  std={src_image.std():.1f}")
print(f"Equalized: mean={equalized_image.mean():.1f}  std={equalized_image.std():.1f}")
print(f"Matched  : mean={matched_image.mean():.1f}  std={matched_image.std():.1f}")
```

Expected Output

The output displays three images: Original, Histogram Equalized, and Histogram Matched. It also displays three corresponding histograms. At the end, the mean and standard deviation values of the three images are printed.

Observation

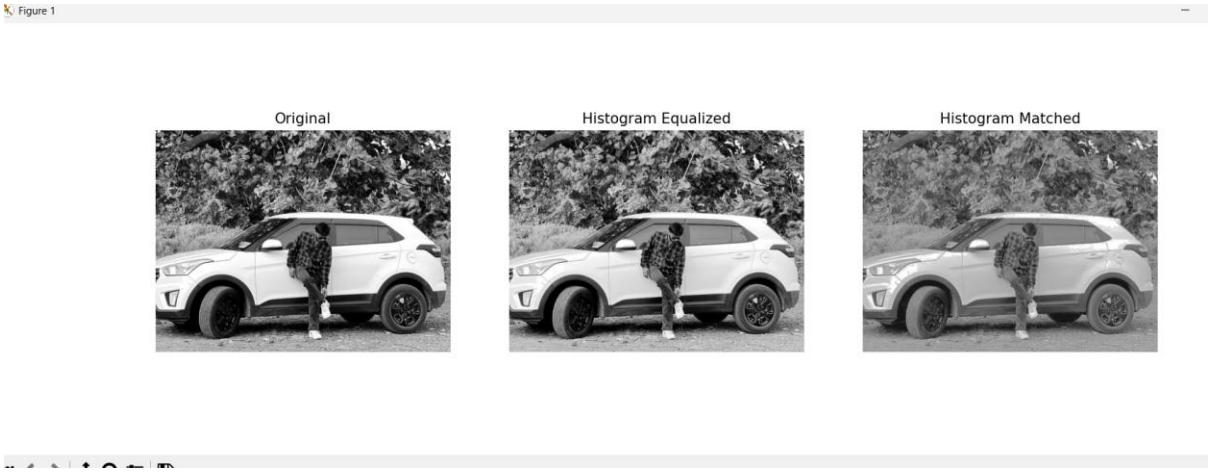
The histogram-equalized image shows improved contrast by redistributing the intensity values. The histogram-matched image has an intensity distribution closer to the selected reference image, giving it a different brightness and contrast appearance. The histograms clearly show the changes in intensity distribution produced by both techniques. The mean and standard deviation values also help compare the brightness and contrast of the original, equalized, and matched images. The histogram-equalized image provides enhanced contrast by spreading the intensity values across the available range. The histogram-matched image

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

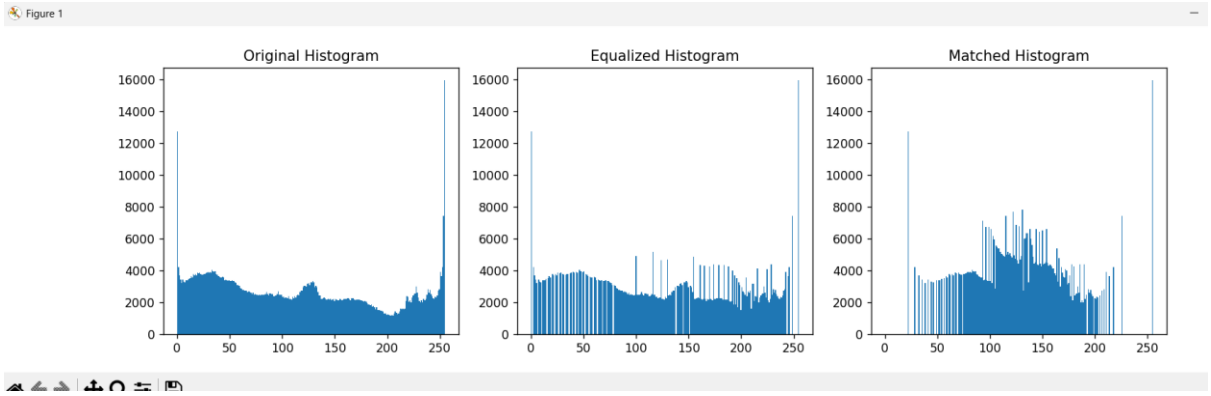
produces an intensity distribution that is more similar to the chosen reference image, resulting in a different level of brightness and contrast. The histograms clearly illustrate the changes in intensity distribution caused by both methods. The mean and standard deviation values also provide a useful numerical comparison of the brightness and contrast of the original, equalized, and matched images.

Output :-

```
PS D:\SEM 3\python> python -u "d:\SEM 3\python\tempCodeRunnerFile.pyt
Original : mean=115.2  std=79.0
Equalized: mean=125.7  std=75.0
Matched  : mean=127.6  std=49.7
```



Histograms:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

Image 3 — Histogram Equalization and Histogram Matching

Explanation

The program reads the image as a grayscale image and checks whether it is loaded correctly. A Gaussian reference image with a mean of 128 and standard deviation of 50 is generated. The original image is enhanced using histogram equalization, while histogram matching changes its intensity distribution according to the generated reference image. The program then displays the original, equalized, and matched images along with their histograms. Finally, the mean and standard deviation of all three images are calculated for comparison.

Code:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

from skimage.exposure import match_histograms


src_image = cv2.imread("ex1_3.png", cv2.IMREAD_GRAYSCALE)


if src_image is None:

    print("Image not found!")

    exit()


np.random.seed(42)

reference_image = np.random.normal(loc=128, scale=50, size=(300, 300))

reference_image = np.clip(reference_image, 0, 255).astype(np.uint8)
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

```
equalized_image = cv2.equalizeHist(src_image)


matched_image = match_histograms(src_image, reference_image)
matched_image = np.uint8(np.clip(matched_image, 0, 255))


plt.figure(figsize=(15, 5))


plt.subplot(1, 3, 1)
plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)
plt.title("Original")
plt.axis('off')


plt.subplot(1, 3, 2)
plt.imshow(equalized_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Equalized")
plt.axis('off')


plt.subplot(1, 3, 3)
plt.imshow(matched_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Matched")
plt.axis('off')
```

```
plt.show()


plt.figure(figsize=(15, 4))


plt.subplot(1, 3, 1)
plt.hist(src_image.ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")


plt.subplot(1, 3, 2)
plt.hist(equalized_image.ravel(), bins=256, range=(0, 255))
plt.title("Equalized Histogram")


plt.subplot(1, 3, 3)
plt.hist(matched_image.ravel(), bins=256, range=(0, 255))
plt.title("Matched Histogram")


plt.show()


print(f"Original : mean={src_image.mean():.1f}  std={src_image.std():.1f}")
print(f"Equalized: mean={equalized_image.mean():.1f}  std={equalized_image.std():.1f}")
print(f"Matched  : mean={matched_image.mean():.1f}  std={matched_image.std():.1f}")
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

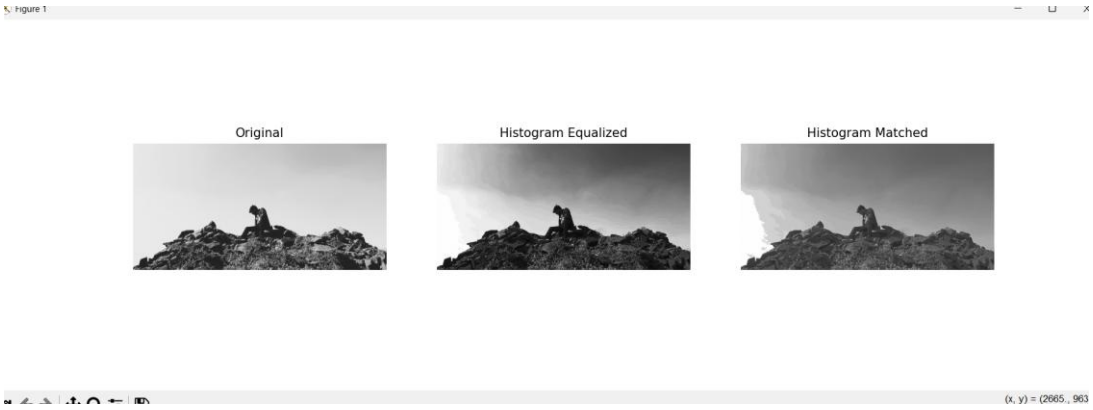
Expected Output

The output displays three images: **Original**, **Histogram Equalized**, and **Histogram Matched**. It also displays the corresponding histograms for all three images. At the end, the program prints the **mean and standard deviation** values of the original, equalized, and matched images.

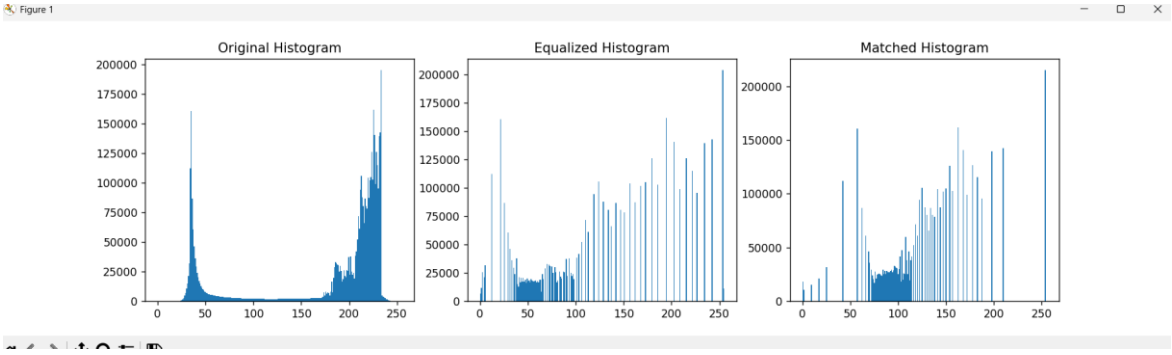
Observation

The histogram-equalized image provides enhanced contrast by spreading the intensity values across the available range. The histogram-matched image produces an intensity distribution that is more similar to the chosen reference image, resulting in a different level of brightness and contrast. The histograms clearly illustrate the changes in intensity distribution caused by both methods. The mean and standard deviation values also provide a useful numerical comparison of the brightness and contrast of the original, equalized, and matched images.

Output :-



Histograms:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

Image 4 — Night Car Photo (Histogram Equalization + Matching)

Explanation

The program reads the image as a grayscale image and checks whether it is loaded correctly. A Gaussian reference image with a mean of 128 and standard deviation of 50 is generated. The original image is enhanced using histogram equalization, while histogram matching changes its intensity distribution according to the generated reference image. The program then displays the original, equalized, and matched images along with their histograms. Finally, the mean and standard deviation of all three images are calculated for comparison.

Code:-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.exposure import match_histograms


src_image = cv2.imread("ex1_4.png", cv2.IMREAD_GRAYSCALE)


if src_image is None:
    print("Image not found!")
    exit()


np.random.seed(42)
reference_image = np.random.normal(loc=128, scale=50, size=(300, 300))
reference_image = np.clip(reference_image, 0, 255).astype(np.uint8)


equalized_image = cv2.equalizeHist(src_image)
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

```
matched_image = match_histograms(src_image, reference_image)
matched_image = np.uint8(np.clip(matched_image, 0, 255))

plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)
plt.title("Original")
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(equalized_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Equalized")
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(matched_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Matched")
plt.axis('off')

plt.show()

plt.figure(figsize=(15, 4))
```

```
plt.subplot(1, 3, 1)
plt.hist(src_image.ravel(), bins=256, range=(0, 255))
plt.title("Original Histogram")

plt.subplot(1, 3, 2)
plt.hist(equalized_image.ravel(), bins=256, range=(0, 255))
plt.title("Equalized Histogram")

plt.subplot(1, 3, 3)
plt.hist(matched_image.ravel(), bins=256, range=(0, 255))
plt.title("Matched Histogram")

plt.show()

print(f"Original : mean={src_image.mean():.1f}  std={src_image.std():.1f}")
print(f"Equalized: mean={equalized_image.mean():.1f}  std={equalized_image.std():.1f}")
print(f"Matched  : mean={matched_image.mean():.1f}  std={matched_image.std():.1f}")
```

Expected Output

The output displays three images: Original, Histogram Equalized, and Histogram Matched. Three corresponding histograms are also displayed. At the end, the program prints the mean and standard deviation values for the original, equalized, and matched images.

Observation

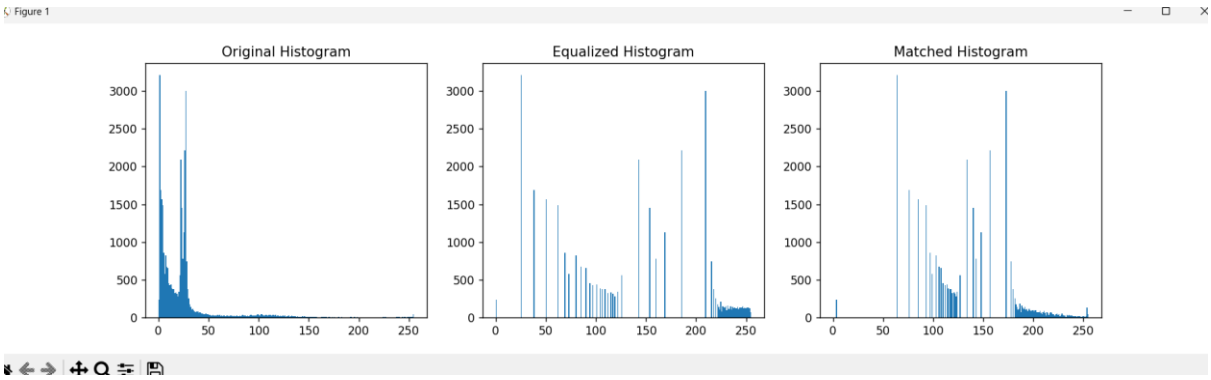
The histogram-equalized image improves the overall contrast by redistributing the pixel intensity values. The histogram-matched image produces an intensity distribution closer to the generated Gaussian reference, resulting in a different brightness and contrast appearance. The histograms clearly show the changes caused by both techniques. The mean

and standard deviation values provide a numerical comparison of the brightness and contrast changes between the original, equalized, and matched images.

Output :-



Histograms:-



Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

Image 5 — Histogram Equalization and Histogram Matching

Explanation

The program reads the image as a grayscale image and checks whether it is loaded correctly. A Gaussian reference image with a mean of 128 and standard deviation of 50 is generated. The original image is enhanced using histogram equalization, while histogram matching changes its intensity distribution according to the generated reference image. The program then displays the original, equalized, and matched images along with their histograms. Finally, the mean and standard deviation of all three images are calculated for comparison.

Code:-

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

from skimage.exposure import match_histograms


src_image = cv2.imread("ex1_5.png", cv2.IMREAD_GRAYSCALE)


if src_image is None:
    print("Image not found!")
    exit()


np.random.seed(42)

reference_image = np.random.normal(loc=128, scale=50, size=(300, 300))

reference_image = np.clip(reference_image, 0, 255).astype(np.uint8)


equalized_image = cv2.equalizeHist(src_image)


matched_image = match_histograms(src_image, reference_image)
```

```
matched_image = np.uint8(np.clip(matched_image, 0, 255))


plt.figure(figsize=(15, 5))


plt.subplot(1, 3, 1)
plt.imshow(src_image, cmap='gray', vmin=0, vmax=255)
plt.title("Original")
plt.axis('off')


plt.subplot(1, 3, 2)
plt.imshow(equalized_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Equalized")
plt.axis('off')


plt.subplot(1, 3, 3)
plt.imshow(matched_image, cmap='gray', vmin=0, vmax=255)
plt.title("Histogram Matched")
plt.axis('off')


plt.show()


plt.figure(figsize=(15, 4))


plt.subplot(1, 3, 1)
plt.hist(src_image.ravel(), bins=256, range=(0, 255))
```

```
plt.title("Original Histogram")


plt.subplot(1, 3, 2)
plt.hist(equalized_image.ravel(), bins=256, range=(0, 255))
plt.title("Equalized Histogram")


plt.subplot(1, 3, 3)
plt.hist(matched_image.ravel(), bins=256, range=(0, 255))
plt.title("Matched Histogram")


plt.show()


print(f"Original : mean={src_image.mean():.1f}  std={src_image.std():.1f}")
print(f"Equalized: mean={equalized_image.mean():.1f}  std={equalized_image.std():.1f}")
print(f"Matched  : mean={matched_image.mean():.1f}  std={matched_image.std():.1f}")
```

Expected Output

The output displays three images: Original, Histogram Equalized, and Histogram Matched. Three corresponding histograms are also displayed. At the end, the program prints the mean and standard deviation values for the original, equalized, and matched images.

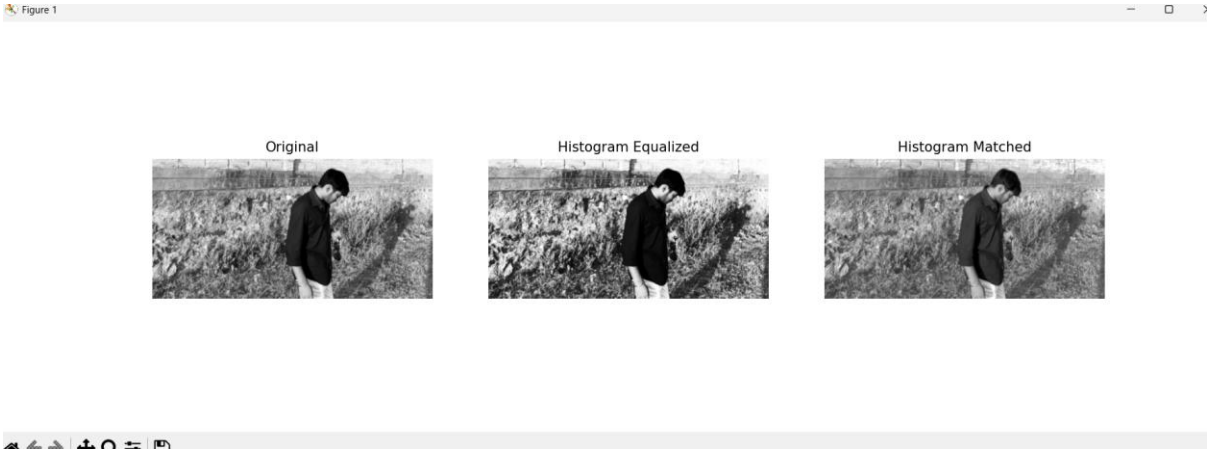
Observation

The histogram-equalized image improves the overall contrast by redistributing the pixel intensity values. The histogram-matched image produces an intensity distribution closer to the generated Gaussian reference image, resulting in a different brightness and contrast appearance. The histograms clearly show the changes produced by both techniques. The mean and standard deviation values provide a numerical comparison of the brightness and contrast changes between the original, equalized, and matched images.

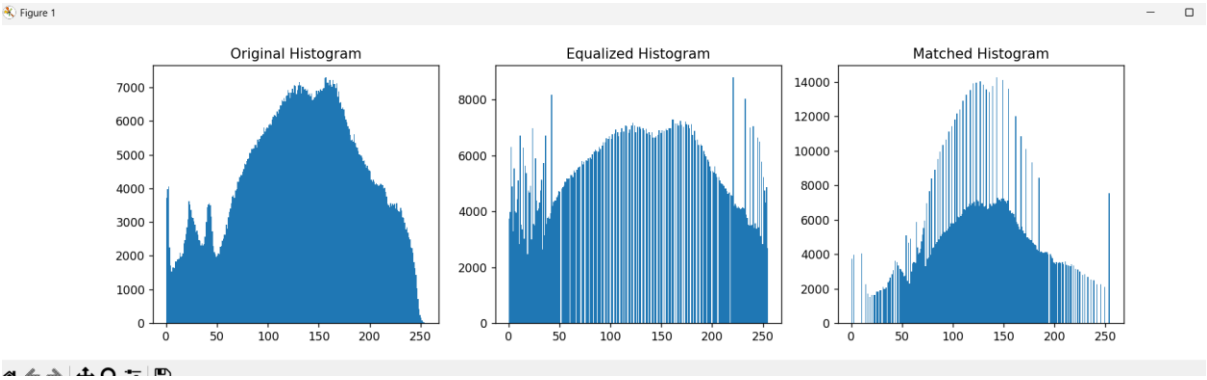
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**

Output :-

Result:-



Histograms:-



Conclusion :-

This experiment successfully demonstrated histogram equalization and histogram matching on five degraded grayscale images. Histogram equalization enhanced the overall contrast by redistributing the pixel intensity values, while histogram matching adjusted the intensity distribution based on the chosen reference image. The histograms clearly displayed the changes in pixel intensity after processing. The mean and standard deviation values were also useful for comparing the brightness and contrast of the processed images. Overall, histogram equalization was effective for general contrast improvement, whereas histogram matching offered more controlled enhancement according to the reference image.

	Marwadi University **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:To enhance the given degraded images using histogram equalization and histogram matching, and compare their effects on image contrast and clarity.**	
Experiment:7	**Date:**	**Enrollment No:92400133079**